



What is Wi-Fi?

- Wireless technology to connect devices to the internet
- Uses radio waves instead of cables
- Commonly used in homes, offices, and public places
- Connects devices like mobiles, laptops, and smart TVs

What Does Wi-Fi Security Mean?

- Protecting a Wi-Fi network from unauthorized access
- Keeps user data safe from hackers
- Uses encryption to secure communication
- Prevents data theft and misuse

What is WEP & Why Is It Insecure?

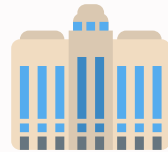
- WEP stands for Wired Equivalent Privacy
- Old and outdated Wi-Fi security protocol
- Uses weak encryption algorithms
- Can be cracked easily in minutes
- No longer considered secure

WPA, WPA2, WPA3 **(What Are They & Key Differences)**

- WPA (Wi-Fi Protected Access): Improved security over WEP
- WPA2: Strong encryption (AES), widely used and more secure
- WPA3: Latest version with enhanced protection
- Security improves from WPA → WPA2 → WPA3
- WPA3 protects better against brute-force attacks



PSK vs



Enterprise

PSK (Pre-Shared Key):

- Single password for all users
- Common in home Wi-Fi networks

Enterprise:

- Individual username and password
- Used in offices and organizations
- More secure and manageable

Real-Life **Wi-Fi Security Mistakes**

- Using weak or common passwords
- Keeping default router credentials
- Enabling WPS unnecessarily
- Not updating router firmware
- Connecting to open public Wi-Fi networks